

**2018 RVHS Year 6 Preliminary Examinations
P4 Confidential Instructions**

Practical Exam	Apparatus (per student)		Materials	
<u>Question 1</u>	2	12 cm ³ syringes	~50 cm ³	Milk labelled M
	2	6 cm ³ syringes	~60 cm ³	Distilled water labelled W
	1	500 cm ³ beaker labelled Waste	30 cm ³	0.1% turmeric solution labelled I
	8	Paper towels	40 cm ³	0.3% sodium carbonate solution labelled A
	6	Medium plastic vials	50 cm ³	5% lipase solution labelled E
	6	Test tubes		
	1	Test tube rack		
	1	Stopwatch		
	1	Marker		
<u>Question 2</u>	4x	Pasteur pipettes	1 cm ³	HCl in each microfuge tube labelled P1, P3, T
	1x	Microtiter plate	1 cm ³	Distilled water in each microfuge tube labelled P2, W, A
	1x	Permanent marker		Universal indicator in a microfuge tube labelled X
	1x	Microfuge tube rack	1 cm ³	
<u>Question 3</u>	1x	Ruler	1 x	Microscope slide labelled S1 (2 students to 1)
	1x	Microscope (2 Students to 1)		

Preparation of solutions and reagents

M, at least 40 cm³ of whole milk in a beaker or container, labelled **M**.
The milk must be full fat (no fat removed) e.g. cows' or goats' milk.
This is sufficient for 1 candidate.

W, at least 50 cm³ of distilled water in a beaker or container, labelled **W**.
This is sufficient for 1 candidate.

I, at least 30 cm³ of 0.1% turmeric solution in a beaker or container, labelled **I**.

This is prepared by putting 0.1 g of turmeric (ground turmeric spice as used in cooking) in 5 cm³ of 70% ethanol (industrial methylated spirits) in a beaker or container, and stirring to dissolve for 5 minutes. Make up to 100 cm³ with distilled water. (Ignore residue left in the bottom.)

Turmeric solution may lose its colour with time. It must be prepared **within one hour** of use by candidates, including those candidates who start with **Question 2**.

This is sufficient for 3 candidates.

A, at least 40 cm³ of 0.3% sodium carbonate solution in a beaker or container, labelled **A**.

This is prepared by putting 0.3 g of anhydrous sodium carbonate in 80 cm³ of distilled water in a beaker and stirring to dissolve. Make up to 100 cm³ with distilled water.

This is sufficient for 2 candidates.

E, at least 50 cm³ of 5% Lipase solution (supplied by Cambridge) in a beaker or container, labelled **E**.

This is prepared by putting 5 cm³ of the Lipase enzyme solution into 70 cm³ of distilled water in a beaker while stirring. Make up to 100 cm³ with distilled water. It must be prepared **within one hour** of use by candidates, including those candidates who start with **Question 2**.

This is sufficient for 2 candidates.